The Geometric Siphon

Emergent Capital Reallocation in Concentrated Liquidity Portfolios

K.R. Ryan

Independent Researcher | ORCID - 0009-0004-6295-7040 | gancor@gancor.xyz

March 2026

Keywords geometric residual · automated market maker · portfolio rebalancing
Uniswap V3 · Aerodrome · DeFi mechanism design

JEL: G12, G14, G19, G23

∴

Abstract. This paper identifies and formalises the *Geometric Siphon*, a previously undocumented mechanism in concentrated liquidity portfolio management. When multiple CL positions share a depositor-level token balance and are autonomously rebalanced, token ratio mismatches between old and new tick ranges produce residuals that flow through the shared balance, transferring capital between positions in independent pools. Three theorems characterise the mechanism. I derive the geometric residual from the Uniswap V3 amount equations. Theorem 1 establishes that the residual is zero if and only if the old and new ranges demand the same token ratio at the current price. Theorem 2 proves that single-pool positions sharing a contract balance converge to equal value under repeated rebalancing. Theorem 3 proves that positions rebalancing without swap correction decay geometrically. A graph-theoretic *Connector Rule* ties per-pool flow direction to portfolio topology, supported by a token level decomposition across ten pools and five depositor groups. A Foundry suite verifies all three theorems against unmodified V3 contracts on Base mainnet, returning a 2,061 USDC residual on range changes against an analytical zero on same-range rebalances. A 1,380-event dataset across 39 positions and seven days on Aerodrome (Base) decomposes observed flow into residual and slippage at Spearman correlation 0.976, revealing a ~5%/~95% split between geometric creation and dust redistribution. A controlled value-to-pool liquidity sweep collapses a 1:3:9 size ratio to 1.37:1:1.20 within 33 events. A USDC/ZARP position loses 99.8% of its capital, falling from $942 to $1.59 across 14 zero-swap rebalances over 65 hours, matching the 11-rebalance extinction bound.

Note on supersession. This preprint and its companion *The Geometric Siphon II: Directional Properties* have been consolidated into a single revised manuscript (*The Geometric Siphon: Existence, Equilibrium, and Directional Properties of the Residual in Concentrated Liquidity Portfolios*). The Foundry verification suite and the public dataset now align with the consolidated manuscript.[1] The empirical findings reported in this preprint remain the citable record for this version of the analysis.

[1]Consolidated manuscript: `https://github.com/g4nc0r/the-geometric-siphon`. Foundry suite (Code DOI): `https://doi.org/10.5281/zenodo.19905208`. Public dataset (Data DOI): `https://doi.org/10.5281/zenodo.19904832`.

1. Introduction

Concentrated liquidity, introduced by Uniswap V3,[1] requires active management. Every major CL automation platform (Gamma Strategies, Arrakis Finance, Beefy CLM) uses a *vault-per-pool architecture* in which one vault manages one token pair and leftover tokens from rebalances are reinvested into the same position.

```
Vault-per-pool:    dustBalance[vault]              -> per-pool isolation
Position Manager:  dustBalance[depositor][token]   -> cross-pool sharing
```

Position Manager V6 (PM V6), a CL management contract on Aerodrome (Base), takes a different approach. Leftover tokens sit in `dustBalance[depositor][token]`, keyed by depositor and token with no pool dimension. Every position under one depositor draws from the same balance. A capital efficiency decision that unintentionally created cross-position capital flow.

On 2 March 2026, routine monitoring revealed a USDC/CHECK position had grown ~65% whilst a USDC/ODOS position in the same depositor group had shrunk ~60%. They shared no tokens except USDC. CHECK was consuming USDC residuals left behind by ODOS rebalances through the shared dust pool.

The initial assumption was slippage, but on-chain decomposition of 92 rebalance transactions showed that the transfers are driven almost entirely by the geometric mismatch between consecutive tick ranges. I term this the *Geometric Siphon*.

Table 1. Phase overview.

Phase	Period	Positions	Events	Contract	Focus
1	3–5 Mar 2026	22	738	PM V6	Discovery, topology, regime effects
2	7–9 Mar 2026	17	642	PM V7	Formalisation, equilibrium, Foundry proof
Total	**7 days**	**39**	**1,380**	—	—

Theorem 1 derives a closed-form residual fraction $\Delta R/V$ from the Uniswap V3 amount equations, with the residual vanishing if and only if the old and new ranges demand the same token ratio at the current price; Foundry tests against unmodified V3 contracts confirm that range changes create residuals, same-range rebalances do not, and residuals scale linearly with position size (§6). Theorem 2 proves that single-pool positions sharing a contract balance converge to equal value under repeated rebalancing (§3.4), confirmed empirically by a V/L_p sweep in which a 1:3:9 size ratio collapses to 1.37:1:1.20 within 33 events (§5.3). Theorem 3 proves that positions rebalancing without swaps decay geometrically, confirmed by a complete USDC/ZARP lifecycle from \$942 to \$1.59 in 14 zero-swap rebalances (§5.8) and four additional Foundry tests.

Three further results characterise structure. The Stage 1/Stage 2 split distinguishes geometric creation (~5% of events) from dust pool redistribution (~95% of events), governed by different variables (§3.3). A graph-theoretic *Connector Rule* identifies the per-pool driver as the token with higher portfolio connectivity rather than the more volatile token (§3.5, §5.2). A Simpson's paradox in CL analysis emerges from V/L_p and volatility σ being complementary predictors that reverse sign on aggregation (§5.2). Three directional properties of the residual (monotonicity in displacement, directional asymmetry in USD terms, exit-side asymmetry) are derived and verified as Theorems 4–6 in the companion paper Ryan,[2] sharing the same Foundry verification suite.

2. Related work

Adams *et al.*[1] introduced concentrated liquidity in Uniswap V3. Fan *et al.*[3] formulated the strategic LP problem of choosing tick ranges to maximise fees net of repositioning costs, treating each position in isolation. This paper studies what happens when positions share depositor-level balances.

Heimbach *et al.*[4] model and empirically analyse V3 LP risks and returns, finding wide cross-strategy variation that rewards active management. This paper isolates the residual from fee accrual and impermanent loss by taking the rebalancing decision as given.

Hasbrouck *et al.*[5] model V3 CL provision as a covered call sold at intrinsic rather than market value, so LPs forgo the option's time premium in exchange for fees, and equilibrium liquidity provision decreases in that time premium. The geometric residual is a separate loss arising at the rebalance itself, determined by the geometry of $R(s, s_a, s_b)$ rather than by option-theoretic exposure between rebalances.

Milionis *et al.*[6] introduced LVR (Loss-Versus-Rebalancing), the return gap between an AMM LP and a continuously rebalancing benchmark, driven by arbitrageurs trading against stale AMM prices. Milionis *et al.*[7] extend the framework with trading fees and discrete block timing. LVR is a continuous holding cost; the geometric residual is a discrete event cost at rebalance, determined by the V3 amount equations rather than by arbitrage. The decomposition in §5.1 attributes over 97% of observed flow to the residual rather than to swap-side friction.

Cartea *et al.*[8] introduce *predictable loss*, a closed-form measure of the unhedgeable losses an LP incurs from holding assets in a CL pool, comprising convexity loss from informed trading and opportunity cost on locked capital. Predictable loss accrues between rebalances; the geometric residual arises at the rebalance itself, depends on range geometry rather than price trajectory, and is zero on same-range rebalances regardless of price travel (Theorem 1).

Willetts and Harrington[9] measure arbitrage efficiency and rebalancing costs in dynamic weight AMM pools on Ethereum and Base using LVR benchmarks, the closest empirical context for per-rebalance costs on Base L2.

Balancer[10] rebalances via its AMM invariant within a single pool, Set Protocol uses oracle-triggered swaps, and Yearn deploys capital across strategist-managed strategies. All require a shared pool invariant, external price feeds, or manual intervention; the siphon arises from CL geometry alone.

Maverick Protocol[11] introduced directional liquidity modes where bins follow price automatically. Its 'Mode Both' is the closest analogue to siphon-driven repositioning, being continuous, automatic, and oracle-free. Maverick operates within a single position; the siphon operates *across* positions in different pools.

To my knowledge, geometric residual flows between CL positions sharing depositor-level balances have not been previously described.

3. Theoretical framework

3.1. Concentrated liquidity preliminaries

For a position with liquidity L in tick range $[a, b]$ at current sqrt price $s = \sqrt{p}$ (with $s_a \leq s \leq s_b$), the Uniswap V3 amount formulas give:

$$x = L \cdot \left(\frac{1}{s} - \frac{1}{s_b}\right), \quad y = L \cdot (s - s_a) \tag{1}$$

where x is token0 (lower address) and y is token1 (higher address).

The token ratio demanded by range $[s_a, s_b]$ at price s:

$$R(s, s_a, s_b) = \frac{x}{y} = \frac{1/s - 1/s_b}{s - s_a} \tag{2}$$

This ratio depends only on the sqrt price and range boundaries, independent of L. Two positions of different sizes in the same range at the same tick have identical token ratios.

The position value in token1 units:

$$V = x \cdot s^2 + y = L \cdot \phi(s, s_a, s_b) \tag{3}$$

where $\phi(s, s_a, s_b) = 2s - s^2/s_b - s_a$ is the value factor.

Flow notation. L_p denotes the total liquidity of the pool containing the position, distinguished from the position's own L. For a rebalance, ΔR denotes the gross geometric residual predicted by (1) under an isolated withdraw-and-mint sequence, characterised in Theorem 1 below. S denotes the swap slippage absorbed by an internal rebalance swap. $D = \Delta R - S$ denotes the observed net dust flow. Theorem 1 characterises ΔR, and §5 reports D and S empirically.

3.2. The geometric residual

Definition. A rebalance withdraws liquidity from range $[s_a, s_b]$ and mints into range $[s_{a'}, s_{b'}]$, both at the same current price s.

Withdrawn amounts:

$$x_w = L \cdot (1/s - 1/s_b), \quad y_w = L \cdot (s - s_a) \tag{4}$$

The new range demands ratio $R_{\text{new}} = R(s, s_{a'}, s_{b'})$. If $R_{\text{old}} \neq R_{\text{new}}$, the withdrawn tokens cannot be fully deposited: one is the *binding constraint*, the other has a surplus. The new liquidity is the minimum of the per-token capacities,

$$L_{\text{new}} = \min\left(\frac{x_w}{1/s - 1/s_{b'}}, \frac{y_w}{s - s_{a'}} \right), \tag{5}$$

with the token1 factor $(s - s_{a'})^{-1}$ binding when $R_{\text{old}} \geq R_{\text{new}}$ (excess token0) and the token0 factor $(1/s - 1/s_{b'})^{-1}$ binding when $R_{\text{old}} < R_{\text{new}}$ (excess token1). The fraction of original value captured is

$$\frac{V_{\text{new}}}{V_{\text{old}}} = \frac{L_{\text{new}} \cdot \phi(s, s_{a'}, s_{b'})}{L \cdot \phi(s, s_a, s_b)}. \tag{6}$$

The geometric residual fraction (isolated case):

$$\frac{\Delta R}{V} = \frac{V_{\text{new}}}{V_{\text{old}}} - 1, \tag{7}$$

where $\Delta R = V_{\text{new}} - V_{\text{old}}$ is the isolated residual value (equivalently, the observed D when no swap and no shared-balance contribution are present). In a shared-balance context the observed fraction D/V can be positive (the new position absorbs surplus tokens from the shared balance beyond its withdrawal) or negative (the withdrawal's surplus enters the shared balance as dust).

Theorem 1 (Geometric Residual)**.** *For a concentrated liquidity rebalance from range $[s_a, s_b]$ to $[s_{a'}, s_{b'}]$ at current sqrt price s with $s_a \leq s \leq s_b$ and $s_{a'} \leq s \leq s_{b'}$, and with no shared-balance contribution to the mint (so $D = \Delta R$):*

$$\frac{\Delta R}{V} = 0 \iff R(s, s_a, s_b) = R(s, s_{a'}, s_{b'}).$$

Range equality $[s_a, s_b] = [s_{a'}, s_{b'}]$ is a sufficient condition for $R_{old} = R_{new}$; it is not the only one. At fixed s, the locus $\{(s_{a'}, s_{b'}) : R(s, s_{a'}, s_{b'}) = R_{old}\}$ is a smooth one-dimensional curve in the $(s_{a'}, s_{b'})$ plane, and any range change off this curve produces a strictly non-zero ΔR.

Proof. (⇐) Suppose $R_{\text{old}} = R_{\text{new}}$. The withdrawn amounts (x_w, y_w) are in the ratio required by the new range, so both tokens are deposited in full with no surplus; the new liquidity is $L_{\text{new}} = L(s - s_a)/(s - s_{a'})$. Substituting $R_{\text{old}} = R_{\text{new}}$ into the value factor gives $\phi(s, s_{a'}, s_{b'}) = (s - s_{a'})\,\phi(s, s_a, s_b)/(s - s_a)$, so

$$V_{\text{new}} = L_{\text{new}}\,\phi(s, s_{a'}, s_{b'}) = \frac{L(s - s_a)}{s - s_{a'}} \cdot \frac{(s - s_{a'})\,\phi(s, s_a, s_b)}{s - s_a} = L\,\phi(s, s_a, s_b) = V_{\text{old}},$$

and $\Delta R/V = 0$.

(⇒) Suppose $R_{\text{old}} \neq R_{\text{new}}$. By (5), exactly one token binds the new liquidity and the other has a strict surplus exiting the mint as dust of strictly positive token value $|\Delta R|$. By token value conservation, $V_{\text{new}} = V_{\text{old}} - |\Delta R| < V_{\text{old}}$, so $\Delta R/V < 0$.

The ratio-preserving locus $\{(s_{a'}, s_{b'}) : R(s, s_{a'}, s_{b'}) = R_{\text{old}}\}$ is the zero set of $(1/s - 1/s_{b'})(s - s_a) - (1/s - 1/s_b)(s - s_{a'}) = 0$, a single equation in two unknowns which generically defines a smooth one-dimensional curve; the point (s_a, s_b) lies on this curve but does not exhaust it. ■

Corollary (Width and displacement effects)**.** *Write $w = s_b - s_a$ for the full range width. The required token ratio R depends on the range centre and width but not on L, and is strictly decreasing in w. Consequently:*

(i) Width change. *Widening ($w_{new} > w$) gives $R_{new} < R_{old}$ (position produces token0 excess); narrowing gives $R_{new} > R_{old}$ (token1 excess). In a shared-balance context the widening case tends towards net absorption ($D/V > 0$), narrowing towards net donation ($D/V < 0$).*

(ii) Displacement. *For a rebalance recentred on the current price after the tick moves by δ from the old range midpoint, $|\Delta R/V|$ is strictly increasing in $|\delta|$ for $|\delta| \leq w/2$. A detailed proof is given as Theorem 4 in Ryan.[2]*

3.3. Two-mechanism structure

Of 642 Phase 2 events (§5.7), 95% of rebalances do not change the tick range; they withdraw and re-mint at identical boundaries with no swap. The formula correctly predicts $\Delta R = 0$ for these events. Yet they produce a mean $|D| = \$20$ in dust flow, driven entirely by the contract dust pool.

Two distinct mechanisms are at work:

Stage 1, Geometric Creation (the source). Range-change rebalances (~5% of events) produce geometric residuals via equation (5). The position manager's internal swap absorbs ~94% of the token ratio mismatch; the remaining ~6% enters the shared dust pool as unmatched tokens. Without range changes, the dust pool drains to zero.

Stage 2, Dust Pool Redistribution (the transport). Same-range rebalances (~95% of events) interact with accumulated contract dust. The CL mint function uses all available tokens (withdrawn amounts plus contract dust):

$$L_{\text{new}} = \min\left(\frac{x_w + x_{\text{dust}}}{1/s - 1/s_{b'}},\ \frac{y_w + y_{\text{dust}}}{s - s_{a'}}\right). \tag{8}$$

If contract dust provides extra of the non-binding token, the position absorbs beyond its withdrawal ($D > 0$). If the position's withdrawal slightly over-supplies one token (from accrued fees changing the ratio), the excess stays as dust ($D < 0$).

V/L_p governs Stage 2. Low-V/L_p positions withdraw small amounts relative to the dust pool, so contract dust represents a large proportional gain. High-V/L_p positions withdraw large amounts, so the dust pool is a rounding error relative to their size. The relationship is non-linear. Absorption peaks at $V/L_p \approx 0.05$–0.10, crosses zero near $V/L_p^* \approx 0.13$, and donation intensifies monotonically above that threshold (§5.3). Most events (73%) cluster at $V/L_p < 0.01$ where effects are negligible, making the aggregate linear correlation weak ($r \approx -0.1$) despite a clear directional relationship across the full V/L_p range.

The siphon requires all three components:

Table 2. Necessary components of the siphon mechanism.

Remove…	Effect	Siphon?
Stage 1 (no range changes)	No dust created	Stops: pool drains to zero
Stage 2 (no same-range rebals)	Dust accumulates, never absorbed	Stops: stranded tokens
Shared balances	Each position's dust isolated	Stops: no transfer medium

In steady state, geometric residuals from range changes feed the pool at the same rate that same-range rebalances drain it.

3.4. Equilibrium

Theorem 2 (Single-Pool Convergence)**.** *N positions in the same pool and range, sharing a contract balance, converge to equal value under repeated rebalancing.*

Proof. Setup. Positions $V_1, \ldots, V_N$ in a single CL pool. All share the same tick range, so $L_i \propto V_i$. They share a contract token balance (the dust pool).

Fact 1, Donation scales with size. When position i produces a geometric residual, $|\Delta R_i| \propto L_i \propto V_i$. As a fraction of value, $D_i/V_i = c$ is approximately constant across positions. Larger positions donate more in absolute terms, the same in proportional terms.

Fact 2, Absorption is size-independent. When position i rebalances, the CL mint function (8) uses all available tokens. The dust absorbed is D_{pool}, regardless of V_i. The proportional gain is D_{pool}/V_i, inversely proportional to size.

Growth equation. If each position has equal rebalance probability $\pi = 1/N$ (distinct from the price p of §3.1):

$$\mathbb{E}\left[\frac{\dot{V}_i}{V_i}\right] = \underbrace{\frac{\pi \cdot D_{\text{pool}}}{V_i}}_{\text{absorption} \propto 1/V_i} - \underbrace{c}_{\text{donation rate (constant)}} \tag{9}$$

The first term is strictly decreasing in V_i. The second is constant. Smaller positions grow faster proportionally.

Fixed point. At equilibrium, $\dot{V}_i/V_i = \dot{V}_j/V_j$ for all i, j:

$$\frac{\pi \cdot D_{\text{pool}}}{V_i} = \frac{\pi \cdot D_{\text{pool}}}{V_j} \implies V_i = V_j \tag{10}$$

Equal value is the unique stable equilibrium. ∎

Convergence rate. The growth rate gap between the smallest and largest position:

$$\frac{\dot{V}_s}{V_s} - \frac{\dot{V}_l}{V_l} = \pi \cdot D_{\text{pool}} \cdot \left(\frac{1}{V_s} - \frac{1}{V_l}\right) \tag{11}$$

The gap is positive whenever $V_s < V_l$ and shrinks as the gap closes, giving exponential convergence that decelerates near equilibrium.

Multi-pool extension. For positions across pools with total liquidities $L_{\text{pool},k}$, the equilibrium condition generalises:

$$\frac{V_i}{L_{\text{pool},k(i)}} = \frac{V_j}{L_{\text{pool},k(j)}} \quad \forall\ i,j \tag{12}$$

Different pools have different L_{pool}, so equal V/L_p means different equilibrium sizes. Positions in deeper pools settle at larger values, and hub-spoke topologies are stable because the deep hub sustains a large position whilst volatile spokes sustain smaller ones.

3.5. The Connector Rule

Token level decomposition (§5.2) shows that the siphon driver per pool is determined by *token connectivity* in the portfolio graph.

Definition. For a portfolio with positions in pools $\{P_k\}$, the *connectivity* of token T is the number of pools containing T:

$$\text{conn}(T) = |\{k : T \in P_k\}| \tag{13}$$

The Connector Rule. In pool $P_k = (T_a, T_b)$, the siphon driver is the token with higher connectivity: $\arg\max_{T\in\{T_a,T_b\}} \text{conn}(T)$.

Empirical demonstration. The pool WETH/USDC appears in three different depositor groups with different portfolio topologies:

Table 3. Connector rule across groups.

Group	Portfolio structure	WETH conn	USDC conn	Driver	r
A	WETH hub (3 pools)	3	1	WETH	−0.98
B	USDC hub (4 pools)	3	4	USDC	−0.56
D	Mixed (2 pools)	2	1	USDC	−0.68

The same pool exhibits different driver tokens depending on which of its tokens has higher connectivity in the rest of the portfolio. When WETH is the connector (A), WETH geometric mismatches drive the flow. When USDC is the connector (B), USDC drives it.

Mechanism. When the connector token's price moves, every pool containing that token experiences a correlated geometric mismatch in the connector leg. The non-connector token responds to its own independent dynamics. Correlated mismatches in the connector token create systematic net flows; uncorrelated mismatches in peripheral tokens average towards zero.

3.6. Zero-swap extinction

Theorems 1 and 2 assume the position manager can execute a swap to correct the token ratio after each range change. When a pool has no swap routing ($S = 0$ on every event), the full geometric residual leaks as dust on every rebalance.

Theorem 3 (Zero-Swap Extinction)**.** *Let P be a CL position with value $V_0 > 0$. If the current price exits the position's range and a rebalance is performed with $S = 0$ (no swap), minting into a new range containing the current price, then:*

(i) $V_1 < V_0$ (single-step value loss).

(ii) The residual fraction $\alpha = (V_0 - V_1)/V_0 > 0$ increases monotonically with displacement $|\delta|$ relative to range width w.

(iii) After K consecutive zero-swap rebalances,

$$V_K = V_0 \prod_{k=1}^{K} (1 - \alpha_k),$$

with bounds

$$V_K \leq V_0 (1 - \bar{\alpha}_K)^K, \qquad V_K \leq V_0 (1 - \alpha_{\min})^K,$$

where $\bar{\alpha}_K = \frac{1}{K} \sum_{k=1}^{K} \alpha_k$ *is the mean residual fraction and* $\alpha_{\min} = \min_k \alpha_k > 0$.

(iv) The position reaches any $\varepsilon > 0$ *in at most*

$$K^* = \left\lceil \frac{\ln(V_0/\varepsilon)}{\ln(1/(1 - \bar{\alpha}))} \right\rceil, \qquad K^*_{loose} = \left\lceil \frac{\ln(V_0/\varepsilon)}{\ln(1/(1 - \alpha_{\min}))} \right\rceil.$$

Proof. The proof uses the CL amount equations (1), the mint constraint (8), the rebalance trigger condition, and the finiteness of the shared dust pool.

Part (i). When the price exits above range ($s > s_b$), the withdrawal yields $x_w = 0$ and $y_w = L(s_b - s_a)$; the position is entirely token1. The new range $[s_{a'}, s_{b'}]$ contains s, requiring both tokens. From (8), in isolation ($x_{\text{dust}} = 0$):

$$L_{\text{new}} = \min\left(\frac{0}{1/s - 1/s_{b'}}, \frac{y_w}{s - s_{a'}} \right) = 0, \tag{14}$$

so the entire withdrawn value becomes residual ($\alpha = 1$). In a shared-balance context, contract dust $x_{\text{dust}} > 0$ from other positions may partially compensate the missing token, yielding $L_{\text{new}} > 0$ via (8). Since $x_w = 0$ strictly whilst the new range requires $x > 0$ for $L_{\text{new}} > 0$, and the shared dust pool is finite, the compensation is always partial: $V_1 < V_0$ with $0 < \alpha \leq 1$. The $s < s_a$ case is symmetric at the residual fraction level; USD-denominated values for above-exits versus below-exits on volatile/stablecoin pairs are asymmetric, characterised as Theorem 6 in Ryan.[2]

Part (ii). Let δ denote the displacement of s from the old range midpoint, so the premise of Part (i) requires $|\delta| \geq w/2$. As $|\delta|$ increases, the withdrawn amount of the binding token (opposite to the direction of exit) stays at zero whilst the new recentred range's required token fraction on that side grows with the gap, so α increases monotonically in $|\delta|/w$ for fixed dust supply. A detailed proof of the in-range analogue ($|\delta| \leq w/2$) is given as Theorem 4 in Ryan.[2]

Part (iii). At each rebalance k, Part (i) gives $V_{k+1} = V_k(1 - \alpha_k)$ with $\alpha_k > 0$. By induction,

$$V_K = V_0 \prod_{k=1}^{K} (1 - \alpha_k).$$

Applying AM–GM to the positive terms $\{1 - \alpha_k\}$ yields the tight bound,

$$\prod_{k=1}^{K} (1 - \alpha_k) \leq \left(\frac{1}{K} \sum_{k=1}^{K} (1 - \alpha_k) \right)^K = (1 - \bar{\alpha}_K)^K.$$

The loose bound follows from $1 - \alpha_k \leq 1 - \alpha_{\min}$ for every k.

Part (iv). Solving $V_0(1 - \bar{\alpha})^{K^*} \leq \varepsilon$ for K^* and taking the ceiling gives the tight extinction time; the same with $\alpha_{\min}$ in place of $\bar{\alpha}$ gives the loose bound. ∎

Corollary (Irreversibility)**.** *Under sustained $S = 0$ conditions with stable or declining pool liquidity L_p, the value sequence $V_0, V_1, \ldots$ is monotonically decreasing. Recovery requires a change in L_p sufficient to reduce V/L_p below the absorption crossover ($V/L_p^* \approx 0.13$, §5.2).*

Relationship to Theorem 2. Theorem 2 describes the attractor; Theorem 3 describes the trajectory outside it when the self-correcting mechanism (swaps) is absent. In pools with swap routing, the position manager's internal swap absorbs ~94% of ΔR, leaving only ~6% as net dust, which keeps positions within Theorem 2's convergence basin. When $S = 0$, this damping is absent. The full geometric residual flows into dust on every event, and the position decays at the rate predicted by Theorem 3.

4. Experimental design

4.1. Infrastructure

Both phases used the same adaptive rebalancing engine (`cl-manager`). The engine monitors pool swap events via WebSocket, computes rolling volatility σ at 1h/4h/24h windows, and triggers a rebalance when the current tick exits the position's range. New range width is $w = C \cdot \sigma \cdot \sqrt{t}$ (configurable multiplier $C = 2$–4, centred on current tick). Each rebalance is an atomic `rebalanceCL` call: withdraw → optional swap → mint → stake. All rebalances are autonomous with no human intervention between deployment and data collection.

4.2. Phase 1: Mechanism discovery

Table 4. Phase 1 overview.

Parameter	Value
Period	3–5 March 2026, 41.9 hours
Contract	PM V6, Aerodrome (Base)
Positions	22 across 5 depositor groups
Events logged	738 diffusion events, 1,192 rebalances
Termination	Contract exploit (5 Mar) corrupted depositor state

Table 5. Phase 1 group design.

Group	Positions	Design	Key variable
A	3	Similar V/L_p, different volatility	σ isolation
B	7	Pre-existing, CL1/CL10/CL100 mix, 5 shared tokens	Diversity
C	2	Same pool (EURC/USDC CL1), isolated address	Same-pool control
D	5	\$500 each, identical policy, 30× TVL spread	Pool TVL
E	5	Matched TVL (~\$190–323K), 183× emission spread	Emission rate

4.3. Phase 2: Formalisation and verification

Table 6. Phase 2 overview.

Parameter	Value
Period	7 March 2026 – ongoing
Contract	PM V7, Aerodrome (Base)
Positions	17 across 5 depositor groups
Events logged	642 diffusion events

Table 7. Phase 2 group design.

Group	Positions	Design	Key variable
A	3	WETH/USDC hub + CLAWD/USDC + WETH/AERO	Topology
B	4	WETH/USDC hub + WETH/VVV + WETH/AERO + USDC/CHECK	Cross-pool flow
C	5	EURC/USDC + 4 exotic FX pairs (all USDC-paired)	$\sigma \approx 0$ isolation
D	2	WETH/USDC + WETH/EURC	Connector test
E	3	Same pool (KellyClaude/USDC), ratio 1:3:9	Single-pool equilibrium

Three Phase 2 design choices are central. A deliberate hub-spoke topology places WETH/USDC CL50 as the central hub, with every spoke sharing WETH or USDC. A V/L_p sweep experiment deploys three positions at 1:3:9 size ratio in the same pool, isolating position size as the sole variable. A fiat-FX weekend experiment deploys five stablecoin FX pairs on a Sunday with markets closed, providing $\sigma \approx 0$ conditions that isolate the V/L_p effect from volatility.

4.4. Hypotheses

Table 8. Hypotheses for the controlled phases.

ID	Hypothesis	Source	Result
H1	Flow is geometric, not frictional	Phase 1	Yes, $\rho(\Delta R, D) = 0.976$
H2	V/L_p and σ are complementary	Phase 1	Yes, Simpson's paradox confirmed
H3	Same-pool positions converge	Theory (Thm 2)	Yes, 1:3:9 $\rightarrow$ 1.37:1:1.20
H4	Multi-pool positions settle at equal V/L_p	Theory (12)	Yes, hub-spoke stable
H5	Siphon driver is the connector token	Theory (§3.5)	Yes, 10-pool cross-group
H6	Same-range rebalances yield $\Delta R = 0$	Theory (Thm 1)	Yes, 611/611 confirmed
H7	Emission rate correlates with role	Phase 1	Partially supported
H8	$\sigma = 0$ isolates V/L_p as sole predictor	Phase 2 design	Yes, fiat FX confirms

5. Results

5.1. Core decomposition: $D = \Delta R - S$

Swap event logs from 92 on-chain rebalance transactions (Phase 1) were decoded to extract actual swap amounts.

Table 9. Core decomposition of observed dust flow.

Metric	Value	N
Spearman $\rho(\Delta R, D)$	0.976	72
Spearman $\rho(\lvert\Delta R\rvert, \lvert D\rvert)$	0.949	72
Events with swap	72	—
Events without swap	20	—
Mean slippage S	\$1.49	72
Median swap price impact	0.35%	72

The geometric residual ΔR accounts for over 97% of observed flow. Median swap price impact is 0.35% (\$0.41 on a median \$102 swap). Across all 738 Phase 1 events, 536 (73%) involved on-chain swaps, accounting for 97% of gross dollar flow.

5.2. *Predictive analysis and the Simpson's paradox*

Aggregate vs stratified correlations

Pooling all 545 events (with $|D| > \$0.50$) gives $\rho = 0.329$ for V/L_p and $\rho = -0.367$ for σ, suggesting volatility *reduces* flow size. This is a Simpson's paradox.

Stratified by portfolio group:

Table 10. Stratified correlations of $|D|$ with V/L_p and σ_{1h}, by group.

Group	N	$V/L_p \to \lvert D\rvert$	$\sigma_{1h} \to \lvert D\rvert$
A	75	0.054	0.575
B	114	0.562	−0.362
C	55	0.210	−0.082
D	178	0.171	0.086
E	123	0.370	−0.321

V/L_p and σ are complementary: each dominates exactly when the other lacks cross-position variance. B mixes CL1/CL10/CL100 (25× V/L_p spread), so V/L_p captures the signal. A has near-identical V/L_p across positions, so σ takes over.

The aggregate sign reversal occurs because high-σ pools tend to be thin (small TVL), producing smaller absolute dollar flows despite larger relative mismatches. The confound is pool size.

Nonlinear V/L_p transfer function (Phase 2, $N = 606$)

The relationship between V/L_p and D/V is non-linear. Bucketed analysis across all Phase 2 events with V/L_p data:

Table 11. V/L_p transfer function (Phase 2 cross-position scatter).

V/L_p range	N	Mean D/V	Behaviour
< 0.01	445 (73%)	+0.002	Near-zero (position negligible vs pool)
0.01–0.05	98	+0.008	Mild absorption
0.05–0.10	29	+0.065	Peak absorption
0.10–0.20	20	−0.006	Crossover zone
0.20–0.50	9	−0.034	Consistent donation
0.50–1.00	5	−0.069	Massive donation

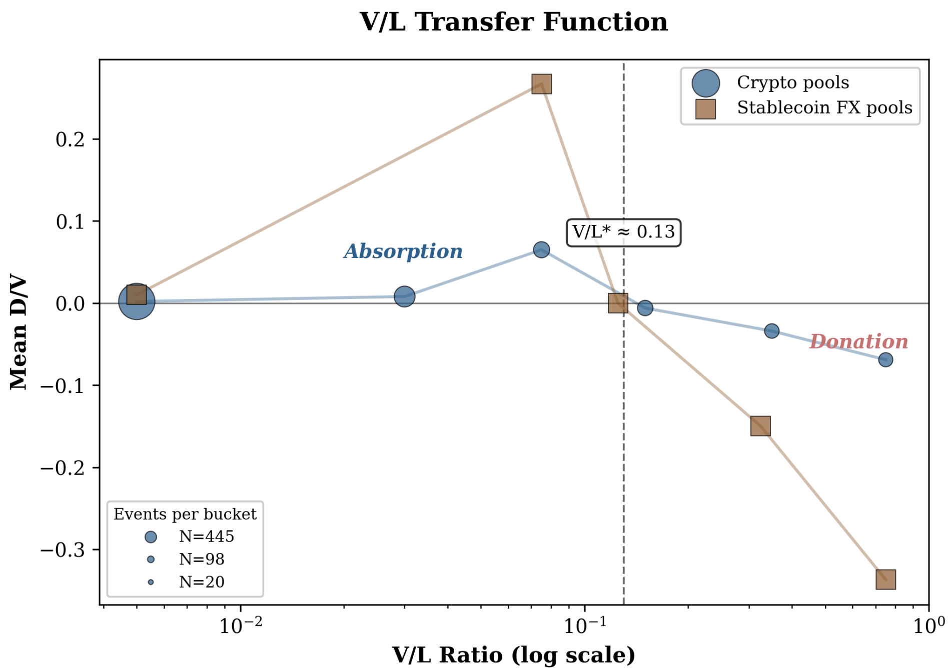

Fig. 1. V/L_p transfer function: absorption peaks at $V/L_p \approx 0.05$–0.10, crosses zero near $V/L_p^* \approx 0.13$, and donation intensifies above. Crypto pools (circles) and stablecoin FX pools (squares) shown; marker size proportional to N.

The non-monotonic shape follows from the two-mechanism structure. At very low V/L_p, positions are too small for contract dust to materially affect them; at moderate V/L_p (0.05–0.10), dust absorption is proportionally large; above the crossover $V/L_p^* \approx 0.13$, positions donate more than they absorb. The 73% of events at $V/L_p < 0.01$ make linear regression misleading ($r \approx -0.1$), but the directional relationship is clear across the full range.

σ amplifies magnitude ($\rho = +0.26$ for σ_{1h} vs $|D/V|$, $N = 536$) but does not determine direction. High σ + high V/L_p compounds donation; high σ + low V/L_p compounds absorption.

Cross-group token decomposition

Token level decomposition of 395 transaction receipts across 10 pools and 5 depositor groups confirms the Connector Rule (§3.5):

Table 12. Cross-group token decomposition.

Pool	Group	Driver token	Driver r	Other r	N
CLAWD/USDC	A	CLAWD	−0.98	−0.42	62
WETH/USDC	A	WETH	−0.98	−0.31	38
WETH/AERO	A	AERO	−0.94	−0.28	31
WETH/VVV	B	WETH	−0.89	−0.15	47
WETH/AERO	B	WETH	−0.97	−0.27	73
WETH/USDC	B	USDC	−0.56	−0.40	25
EURC/USDC	C	USDC	−0.55	—	9
WETH/USDC	D	USDC	−0.68	−0.56	17
KellyClaude/USDC	E	KellyClaude	−0.68	−0.60	20

The WETH/USDC pool appears with three different depositor groups. In A (WETH is the connector), WETH drives at $r = -0.98$. In B (USDC is the connector), USDC drives at $r = -0.56$. The driver changes depending on what other pools share the depositor group.

5.3. Single-pool equilibrium: the V/L_p sweep

Three positions deployed in KellyClaude/USDC CL200 (~\$9,500 TVL) at ratio 1:3:9:

Table 13. V/L_p sweep results after 33 rebalance events.

Position	Deployment	After 33 events	Dust PnL	Role
Small	\$199 ($V/L_p = 0.021$)	~\$900	+\$737	Absorber
Medium	\$597 ($V/L_p = 0.063$)	~\$656	+\$84	Neutral (flipped)
Large	\$1,799 ($V/L_p = 0.189$)	~\$784	−\$933	Donor

Deployment ratio 1:3:9 → ratio at event 33: 1.37:1:1.20. The original size ordering has reversed, and Small is now the largest position.

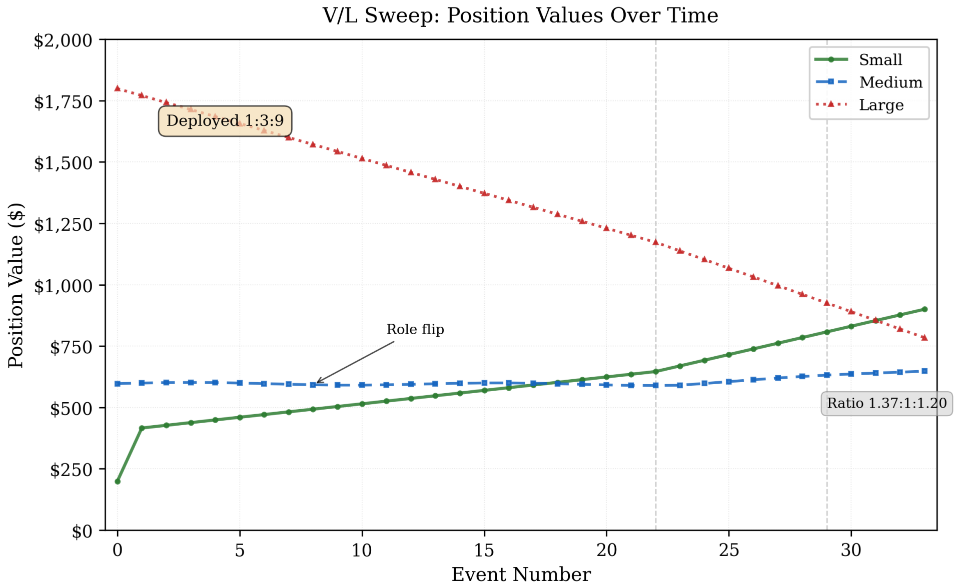

Fig. 2. Single-pool convergence: three positions deployed at ratio 1:3:9 converge towards near-equality over 33 diffusion events. Medium role-flips from donor to absorber at event ~8. Original ordering reverses by event ~28.

Medium role-flipped from initial donor to net absorber after 8 events.

First-rebalance absorption. The first automated rebalance absorbed all contract dust (\$217 from initial tightening) into the Small position, a 114% value increase in one event. The CL mint function uses all available tokens (8), creating winner-take-all dynamics on a per-event basis. Over many events, the probabilistic argument of Theorem 2 prevails, and all positions converge.

Transfer function. Cross-validated with 33 fiat FX events ($N = 55$ total, $r = -0.528$):

Table 14. Cross-validated V/L_p transfer function from the fiat-FX sample.

V/L_p range	Mean D/V	Behaviour
< 0.01	Weak positive	Position too small to absorb meaningfully
0.05–0.10	+0.267	Peak absorption (sweet spot)
0.10–0.15	~0	Crossover zone
0.15–0.50	Negative	Consistent donation
0.50–1.0	−0.337	Massive donation (position dominates pool)

The crossover $V/L_p^* \approx 0.13$–0.15 in liquidity ratio terms. CADC/USDC confirms this independently: pool TVL fluctuated around a fixed-size position, causing V/L_p to traverse 0.003 to 0.32 and the position to role-flip multiple times at the crossover boundary.

5.4. Multi-pool dynamics: hub-spoke topology

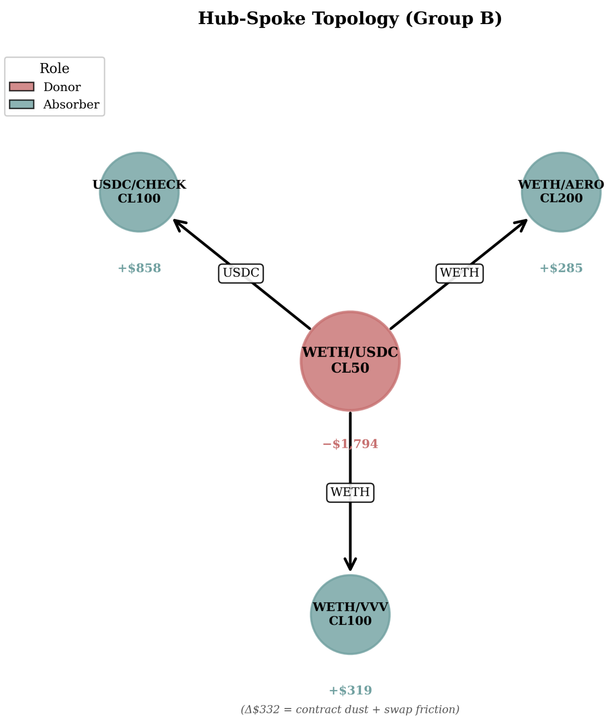

Fig. 3. Hub-spoke topology (Group B): the WETH/USDC hub donates capital through shared WETH and USDC balances to three spoke positions. Δ$332 = contract dust + swap friction.

Phase 2's deliberate hub-spoke design with WETH/USDC CL50 as central hub:
Group B (301 events):

Table 15. Hub-spoke cumulative dust flows in Group B.

Position	Role	Cumulative dust	Events
WETH/USDC CL50 (hub)	Donor	−$1,794	26
USDC/CHECK CL100	Absorber	+$858	153
WETH/AERO CL200	Absorber	+$285	73
WETH/VVV CL100	Absorber	+$319	49

The hub donated \$1,794 and the three spokes collectively absorbed \$1,462. The difference (\$332) is contract dust plus cumulative swap friction.

Temporal causation. Every major hub donation has a matching spoke absorption within 2 hours. The hub leaves WETH residuals in the shared balance, and WETH-paired spokes pick them up on their next rebalance.

Multi-pool stability. Unlike single-pool positions (which equalise), multi-pool positions do not converge to equal value. The hub maintains a larger position because WETH/USDC CL50 has deeper TVL (lower V/L_p at the same capital). Equation (12) predicts this: equal V/L_p across pools means position sizes proportional to pool depth.

5.5. Regime analysis

The 41.9-hour Phase 1 observation spans five distinct market phases:

Table 16. Phase 1 regime partitioning by market state.

Phase	Period (UTC)	Events	Net \$	Character
P1: Stress	03 Mar 17:49 – 04 Mar 06:00	217	−\$196	ETH \$2,215 → \$1,984
P2: Recovery	04 Mar 06:00 – 12:00	137	+\$960	Volatility subsides
P3: Rally	04 Mar 12:00 – 20:45	170	−\$240	BTC → \$71.4K
P4: Post-Rally	04 Mar 20:45 – 05 Mar 04:34	147	−\$323	Consolidation
P5: Calm	05 Mar 04:34 – 11:47	67	+\$86	Diverse portfolios stabilise

Regime-dependent roles. During stress (P1), volatile positions shed capital as geometric mismatches widen. During the rally (P3), the same positions absorb. No position is permanently a donor or absorber.

5.6. Stablecoin FX isolation

Five stablecoin FX pairs deployed on a Sunday (markets closed, $\sigma \approx 0$) provide a natural control experiment isolating V/L_p from volatility:

Table 17. Stablecoin FX isolation: five positions deployed at $\sigma \approx 0$.

Position	V/L_p	σ_{24h}	Cumulative dust	Swaps
ZARP	0.958	98.8	−\$601	0
VCHF	0.034	1.4	+\$72	0
CADC	0.003	5.0	+\$131	0
KRWQ	0.025	1.5	+\$432	0
EURC	0.001	0.1	+\$216	0

Zero swaps across all fiat rebalances (`amountIn = 0` in every `rebalanceCL` call). All dust flow is pure geometric residual ΔR with no swap friction component.

ZARP ($V/L_p \approx 0.96$, meaning the position accounts for nearly all the pool's liquidity) donated a mean 27% of its value per donation event. All five FX pairs were deployed with equal capital. ZARP shrank steadily (V/L_p: 0.958 → 0.974 → 0.992) whilst KRWQ and CADC grew. Capital was conserved at portfolio level.

FX markets were closed and no swaps occurred in any fiat rebalance. Every observed capital flow is therefore attributable to CL geometry alone, confirming V/L_p as the sole predictor when σ is eliminated.

5.7. Formula verification: the two-mechanism split

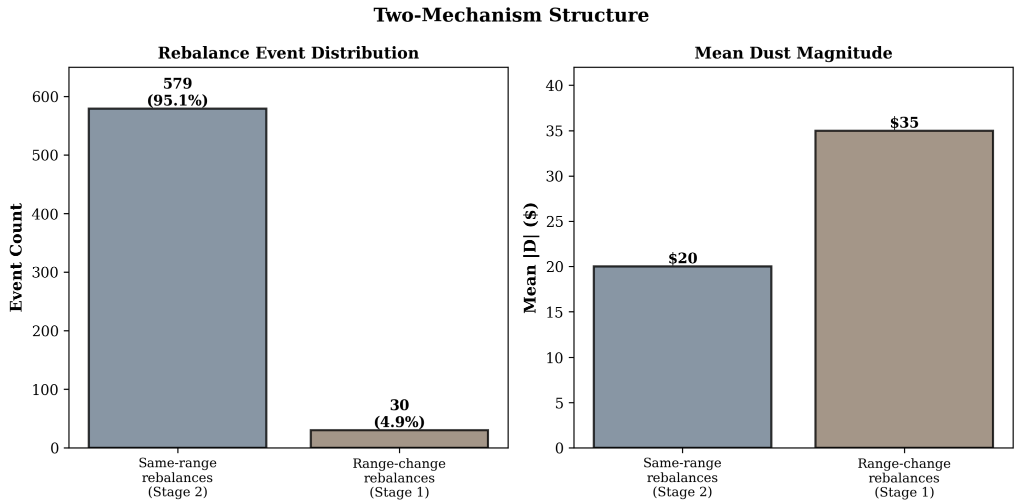

Fig. 4. Two-mechanism structure: 95% of events are same-range rebalances (Stage 2, dust redistribution) with mean $|D|$ = \$20; 5% are range changes (Stage 1, geometric creation) with mean $|D|$ = \$35.

Verification of the closed-form D/V against 642 Phase 2 events:

Table 18. Two-mechanism split: 642 Phase 2 events.

Category	Events	Range changes?	$\Delta R = 0$?	Mean $\lvert D\rvert$
Same-range	611 (95%)	No	Yes (all 611)	\$20
Range-change	31 (5%)	Yes	No (all 31)	\$35

Same-range confirmation. All 611 same-range events have $\Delta R = 0$ as predicted by Theorem 1. Their non-zero dust flows ($\overline{|D|}$ = \$20) are entirely Stage 2 (dust pool redistribution).

Range-change confirmation. All 31 range-change events have $\Delta R \neq 0$. Of the 28 events where the current tick was within the old range (permitting unambiguous case assignment), the formula predicts the direction correctly in 19 cases (70%). Magnitude comparison: predicted $\overline{|\Delta R/V|} = 44\%$, actual $\overline{|D/V|} = 2.7\%$. The internal swap absorbs ~94% of the gross geometric mismatch, leaving ~6% as net dust. The formula describes the gross geometric residual ΔR; the swap partially offsets it but does not eliminate it, so observed $|D/V|$ is smaller than predicted $|\Delta R/V|$.

5.8. Position extinction: the ZARP case study

The USDC/ZARP CL10 position provides a complete empirical lifecycle of Theorem 3, from deployment to extinction, with $S = 0$ on every event.

ZARP (South African Rand stablecoin) had no DEX swap routing. The Odos aggregator returned no path. Every rebalance was a pure withdraw → mint with no token ratio correction, so $D = \Delta R$ exactly. This isolates the geometric residual with no swap confounders.

Table 19. USDC/ZARP CL10 extinction trajectory over 14 rebalance events.

Event	Time (UTC)	Pre ($)	Post ($)	Dust D	V/L_p	α	Pool L_p
1	07 Mar 17:05	942	791	−151	0.958	0.16	135M
2	07 Mar 19:30	791	532	−259	0.974	0.33	223M
3	08 Mar 05:02	532	338	−194	0.992	0.36	693M
4	08 Mar 13:27	338	170	−168	0.960	0.50	144M
5	08 Mar 22:31	170	342	+172	0.790	—	352M
6	09 Mar 00:24	342	171	−171	0.598	0.50	232M
7	09 Mar 04:37	170	51	−119	0.448	0.70	93M
8	09 Mar 04:46	51	281	+230	0.976	—	235M
9	09 Mar 10:18	280	235	−45	0.971	0.16	197M
10	09 Mar 11:08	235	188	−47	0.966	0.20	158M
11	09 Mar 13:47	187	77	−110	0.917	0.59	68M
12	09 Mar 16:40	77	248	+171	0.152	—	1,341M
13	09 Mar 21:28	249	264	+15	0.159	—	1,353M
14	10 Mar 09:52	264	1.59	−262	0.001	0.99	1,137M

Swaps executed: 0 of 14. Total donated: $1,526. Total absorbed: $588 (4 events). Net: −$938 (99.8% of capital).

Three phases. Events 1–4: the position dominated the pool ($V/L_p > 0.95$) and donated on every rebalance. Events 5–8: L_p fluctuated by 4×, producing alternating donation and absorption. Events 9–14: L_p increased 20× between events 11 and 12, reducing V/L_p to 0.001. The final rebalance produced $\alpha = 0.99$.

Extinction bound. Theorem 3(iv) gives both a tight bound based on $\bar{\alpha}$ and a loose bound based on $\alpha_{\min}$. With $\bar{\alpha} = 0.45$ across the 10 donation events,

$$K^* = \left\lceil \frac{\ln(942/1.59)}{\ln(1/0.55)} \right\rceil = \lceil 10.7 \rceil = 11 \text{ rebalances.}$$

Using $\alpha_{\min} = 0.16$ (Event 1, smallest donation), the loose bound gives

$$K^*_{\text{loose}} = \left\lceil \frac{\ln(942/1.59)}{\ln(1/0.84)} \right\rceil = \lceil 36.7 \rceil = 37 \text{ rebalances.}$$

The tight bound predicts 11 rebalances, against 10 donation events observed; the remaining 4 events were absorptions from exogenous liquidity changes and lie outside Theorem 3's pure-decay regime.

Corollary validation. The three largest absorption events (5, 8, 12) each coincided with a substantial increase in pool liquidity L_p. After each absorption, the position returned to donation on the subsequent rebalance, consistent with the irreversibility corollary.

6. Independent verification: Foundry proof

The closed-form derivation is verified with a Foundry test suite covering both Theorem 1 (geometric residual existence and scaling, §6.1) and Theorem 3 (zero-swap extinction, §6.5). Two complementary verification surfaces are used:

Mock-pool tests (`MockCLPool`) implement the Uniswap V3 amount equations against a self-contained pool with a linearised tick-to-sqrt-price approximation. They exercise the residual geometry in isolation, with no fees, no slippage, and no dependence on a live RPC endpoint, but are not Uniswap-accurate in the last significant figure. Fork tests against unmodified Aerodrome Slipstream contracts on Base mainnet, run via Foundry's `vm.createSelectFork`, exercise the same scenarios on real on-chain V3 geometry. They give the canonical demonstration that the residual is a property of unmodified V3 contracts, at the cost of numerical residuals being non-deterministic between runs as the live pool state shifts.

The mock-pool and fork-pool surfaces verify the same theorem at different displacement magnitudes; their qualitative agreement (residual exists, control case is dust-only, residual scales with size) is the central reproducibility claim. §6.1–§6.5 report the mock-pool numbers because they are deterministic and small enough to quote directly; §6.9 reports the fork-pool verification.

6.1. Test 1: Range change creates residual

Table 20. Foundry Test 1: range change creates residual.

Step	Value
Initial position	1 WETH + 2,500 USDC, ±500 ticks around tick 73,135
Price movement	+200 ticks
New range	±1,000 ticks (wider)
Withdrawn	0.076 WETH + 3,434 USDC
Minted	6,681M liquidity (from 23,525M)
Residual	2,061 USDC returned to depositor balance

Assertion: `assertTrue(residual0 > 0 || residual1 > 0)`. PASSED.

6.2. Test 2: Same range = zero residual (control)

Same setup, same price movement, but re-mint into the *identical range*. Residual: 29 wei ($< 10^{-6}$ USD). Pure integer rounding from Solidity's floor division.

Assertion: `assertLt(residual1, 1e3)`. PASSED.
This confirms the theorem's converse: $R_{\text{old}} = R_{\text{new}} \Rightarrow \Delta R = 0$.

6.3. Test 3: Linear scaling with position size

Table 21. Foundry Test 3: linear scaling with position size.

Position size	Residual (raw)	Ratio
0.5 WETH	1,030	1.0×
2.0 WETH	4,122	4.00×

4× position → 4× residual. The $\Delta R/V$ ratio depends on tick displacement and range width but not on L, so the absolute residual scales linearly with position size. Consistent with equation (6) where $V_{\text{new}}/V_{\text{old}}$ is independent of L.

Assertion: `assertGt(largeResidual, smallResidual)`. PASSED.

6.4. What tests 1–3 establish

Three things follow from these tests. The residual appears against an isolated implementation of the Uniswap V3 amount equations with no additional infrastructure, confirming it originates in the CL curve geometry rather than in any contract-level logic. Same-range rebalances produce zero residual (611/611 events in the empirical dataset, see §5.7), confirming the converse of Theorem 1. A 4× position produces a 4× residual (ratio 4.00), so the $\Delta R/V$ formulation is well-defined and independent of L in the range where ϕ is approximately linear.

The mock pool used by tests 1–3 implements a linearised `getSqrtRatioAtTick` approximation around tick 73,135. This is sufficient to demonstrate the existence and scaling of the residual at the order of magnitude reported above, but introduces small rounding artefacts (the 0.5 WETH residual in Test 3 rounds down from a true value of ~ 1,030.5). For Uniswap-accurate quantitative verification of the same theorem, see the fork-pool results in §6.9.

6.5. Test 4: Single-sided exit with no swap (Theorem 3, part i)

Table 22. Foundry Test 4: single-sided exit with no swap.

Step	Value
Initial position	1 WETH + 2,500 USDC, range [72800, 73000]
Price movement	+200 ticks (above upper bound)
Withdrawal	0 WETH, 2,499,999,999 USDC (100% token1)
New range	[73000, 73400] (requires both tokens)
New liquidity	0 (token0 = 0 $\Rightarrow$ min() constraint binds at zero)
Residual	100% of position value

Assertion: `assertTrue(finalValue < initialValue)` . PASSED. $\alpha = 1.0$: a fully single-sided withdrawal with no swap mints zero liquidity. The entire position becomes dust.

6.6. Test 5: Residual scales with displacement (Theorem 3, part ii)

Position deployed with 1,000-tick range. Price displaced past the upper boundary by varying amounts:

Table 23. Foundry Test 5: residual fraction by displacement.

Displacement	α
100 ticks	0%
300 ticks	37%
600 ticks	100%

Assertion: `assertTrue(alpha_300 > alpha_100 && alpha_600 > alpha_300)` . PASSED. Monotonic increase confirmed. The $\alpha = 0\%$ at 100 ticks reflects the mock pool's linear sqrt-price approximation; on real V3 tick math (and on the live fork test in §6.9), this would be small but strictly positive.

6.7. Test 6: Geometric decay over repeated rebalances (Theorem 3, part iii)

Five sequential rebalance cycles with 150-tick price drift and no swap:

Table 24. Foundry Test 6: geometric decay over five cycles.

Cycle	Value (USD)	Decay factor
0	4,999	—
1	3,823	0.76
2	1,565	0.40
3	1,197	0.76
4	490	0.40
5	374	0.76

Assertion: `assertTrue(V[k+1] < V[k])` for all k . PASSED. Total decay: 92% in 5 cycles. The alternating factor (0.76/0.40) reflects the token ratio flip at each step. The two-step product $0.76 \times 0.40 = 0.30$ is the effective decay constant.

6.8. Test 7: Deep out-of-range terminal event (Theorem 3, part ii extreme)

Table 25. Foundry Test 7: deep out-of-range terminal event.

Step	Value
Initial position	±200 ticks around tick 73,135
Price movement	+2,000 ticks past upper bound
New liquidity	0
α	100%

Assertion: `assertTrue(alpha > 95)` . PASSED. At 2,000-tick displacement, the position is entirely single-sided. This mirrors the ZARP terminal event (Event 14, §5.8: $\alpha = 0.99$).

6.9. Live fork verification

The same scenarios are also run as fork tests against the live Aerodrome Slipstream WETH/USDC pool on Base mainnet, calling the unmodified `NonfungiblePositionManager` and pool contracts directly via `vm.createSelectFork`. Five tests reproduce the range-change residual, same-range control, wider-range case, 4× scaling and the architectural precondition (§7.1) against real on-chain V3 geometry.

Fork residuals are smaller than mock-pool residuals (~\$128 vs ~\$2,061) because the fork test swaps a fixed \$1,000 USDC to move the price, shifting the tick by only ~1–3 ticks rather than the mock's hardcoded +200. The precondition test (`test_stockNfpmDoesNotAbsorbDust`) creates two positions in the same range, leaves a residual via the first, then confirms the second's liquidity does not grow on a same-range rebalance. A stock NFPM has no shared dust balance, so no cross-position absorption occurs.

The fork tests do not pin a block number; numerical residuals shift between runs, but the qualitative claims hold deterministically.

6.10. What the full suite establishes

Tests 1–3 prove Theorem 1 (geometric residual existence and scaling). Tests 4–7 prove Theorem 3 (zero-swap extinction: single-step loss, monotonic displacement scaling, geometric decay, terminal events). The five fork tests in §6.9 reproduce Tests 1, 2, 3 and the architectural precondition against unmodified live Slipstream contracts. The full suite runs in well under a second:

```
forge test -vv
Suite result: ok. 7 passed; 0 failed; 0 skipped  (mock-pool)
Suite result: ok. 5 passed; 0 failed; 0 skipped  (fork, Base RPC)
```

The mock-pool environment has no swaps, zero fees, a single position, and no adaptive engine. The only mechanism present is the CL amount calculation itself, which isolates the residual from all confounding factors. The fork environment adds the real protocol contracts but otherwise exercises the same withdraw/mint sequence. Source is in the project's GitHub repository (see Appendix C).

7. Discussion

7.1. The architectural precondition

Every major CL automation platform (Gamma Strategies, Arrakis Finance, Beefy CLM) uses vault-per-pool architecture. Residual tokens stay within the vault for the same pair. The geometric residual arises inside every vault on every range-change rebalance, but it is reinvested into the same position. It manifests as a small capital efficiency loss, indistinguishable from slippage in standard PnL tracking.

The siphon requires `dustBalance[depositor][token]` keyed without a pool dimension, a storage layout that allows residuals from Pool A to be consumed by Pool B. This is an unusual design. The residual has always existed; cross-pool shared balances make it observable as inter-position flow.

Theorem 1 applies to all Uniswap V3-style concentrated liquidity implementations. Cross-position flow requires one additional condition: shared token balances across positions in different pools. Any protocol moving from vault-per-pool to depositor-level multi-pool management will encounter this mechanism.

The fork-mode test in §6.9 verifies this precondition directly: on a stock NFPM (which does not have shared depositor balances), no cross-position absorption occurs.

7.2. Boundary cases

Same-pool oscillation. Two EURC/USDC CL1 positions (Phase 1) produced oscillating flows with consecutive events showing opposite signs 62% of the time, producing pure dissipation with no useful cross-pool reallocation.

Isolated tokens. cbETH/cbBTC (Phase 1) donated −\$2,413.49, but no other position held cbETH. The residual accumulated as unusable contract dust with no destination. For siphon flow to function, every token in the portfolio should appear in at least two positions.

Fat tails. Two of 23 FLOCK rebalances produced 96% of its lifetime donation. The distribution of per-event dust flows is heavy-tailed, consistent with the dependence on tick displacement relative to range width.

7.3. Limitations

Six limitations should be flagged. All data come from one operator's positions on Aerodrome (Base); Theorem 1 guarantees the residual exists on all V3-style implementations, but the empirical dynamics, the two-mechanism split, the V/L_p transfer function, and the equilibrium convergence rate, may differ on other protocols with different fee structures or rebalancing engines.

The observation window is short, seven days spanning stress, recovery, rally, and calm, with no data on behaviour across full market cycles.

The adaptive rebalancing engine creates a circular dependency: σ determines range width, which determines the geometric residual, which determines dust flow, and the measured variables also influence the inputs.

There is no isolated-dust control; a true counterfactual would be identical positions with per-position dust accounting running simultaneously, and the vault-per-pool comparison is a post-hoc reconstruction.

Theorem 3 is proven for the $S = 0$ case; the general case ($0 < S_{\text{eff}} < 1$, where S_{eff} is the fraction of ΔR recovered by the swap) interpolates between Theorem 2 (convergence) and Theorem 3 (extinction), and the value of S_{eff} below which extinction dominates convergence is not yet characterised.

Theorem 2 proves convergence but does not bound the rate; the V/L_p sweep shows rapid initial convergence but the long-run steady state is unobserved.

7.4. *Future work*

Five extensions follow naturally. An economic analysis quantifying the net cost or benefit of the siphon relative to vault-per-pool (ALM) and static LP strategies requires controlled long-duration comparisons.

On emission governance interaction, preliminary evidence from 33 post-flip events suggests ve(3,3) emission rates correlate with siphon role assignment, but separating emission effects from concurrent market regime changes requires observation across multiple epoch cycles.

An adversarial analysis is also open: an attacker who controls rebalance timing across multiple positions may be able to extract value from the siphon, and the feasibility and cost of such an attack are unknown.

Quantitative magnitude prediction is a further open question; the closed-form formula predicts gross residuals, but precise net prediction requires modelling the swap's absorption fraction (~94% in the present data).

Finally, Theorems 2 and 3 describe opposite regimes (convergence with swaps, extinction without), and a unified growth equation $\dot{V}/V = g(V/L_p, \sigma, S_{\text{eff}})$ that continuously interpolates between these regimes as a function of swap efficiency S_{eff} would unify both results and characterise the transition boundary.

8. Conclusion

When concentrated liquidity positions share depositor-level token balances, autonomous rebalancing produces systematic inter-position capital transfers. This paper has formalised the underlying mechanism and validated it across 1,380 rebalance events, 39 positions and two independent contract deployments.

The geometric residual follows from the Uniswap V3 amount equations. Any CL range change that alters the required token ratio at current price produces a non-zero token surplus or deficit (Theorem 1), verified against unmodified Slipstream contracts on Base via a Foundry fork test. Empirically, this residual accounts for over 97% of inter-position flow ($\rho = 0.976$), with swap friction contributing 0.35% median price impact. Range changes (~5% of events) create the residual and feed it into a shared dust pool; same-range rebalances (~95%) redistribute it, governed by V/L_p. Single-pool portfolios converge towards equal value (Theorem 2); multi-pool portfolios form stable hub-spoke topologies where token connectivity determines the flow direction. When the swap mechanism is absent ($S = 0$), the full residual leaks on every rebalance and positions decay geometrically (Theorem 3); a USDC/ZARP position lost 99.8% of its capital (\$942 $\rightarrow$ \$1.59) in 14 zero-swap rebalances over 65 hours, against a tight extinction bound of 11 rebalances at the average leak rate.

The residual has arisen on every CL rebalance that shifts the required token ratio at the current price, since Uniswap V3 launched in May 2021, but vault-per-pool architecture confines it within individual positions. Cross-position flow requires depositor-level shared balances without pool isolation. Any protocol moving towards multi-pool position management will encounter this mechanism.

References

[1] Adams, H., Zinsmeister, N., Salem, M., Keefer, R., Robinson, D. "Uniswap v3 Core." *Uniswap Labs Technical Whitepaper* (2021) `https://uniswap.org/whitepaper-v3.pdf`.

[2] Ryan, K.R. "The Geometric Siphon II: Directional Properties." *SSRN preprint 6481498* (2026) `https://doi.org/10.2139/ssrn.6481498`.

[3] Fan, Z., Marmolejo-Cossío, F.J., Moroz, D., Neuder, M., Rao, R., Parkes, D.C. "Strategic Liquidity Provision in Uniswap v3." *Proceedings of the 5th Conference on Advances in Financial Technologies (AFT 2023), LIPIcs vol. 282, Article 25* (2023) `https://doi.org/10.4230/LIPIcs.AFT.2023.25`.

[4] Heimbach, L., Schertenleib, E., Wattenhofer, R. "Risks and Returns of Uniswap V3 Liquidity Providers." *Proceedings of the 4th ACM Conference on Advances in Financial Technologies (AFT 2022)* (2022) https://doi.org/10.1145/3558535.3559772.

[5] Hasbrouck, J., Rivera, T.J., Saleh, F. "An Economic Model of a Decentralized Exchange with Concentrated Liquidity." *Management Science (forthcoming)* (2025) https://doi.org/10.1287/mnsc.2024.04510.

[6] Milionis, J., Moallemi, C.C., Roughgarden, T., Zhang, A.L. "Automated Market Making and Loss-Versus-Rebalancing." *arXiv preprint 2208.06046* (2022) https://arxiv.org/abs/2208.06046.

[7] Milionis, J., Moallemi, C.C., Roughgarden, T. "Automated Market Making and Arbitrage Profits in the Presence of Fees." *Financial Cryptography and Data Security 2024* (2024) https://arxiv.org/abs/2305.14604.

[8] Cartea, Á., Drissi, F., Monga, M. "Decentralized Finance and Automated Market Making: Predictable Loss and Optimal Liquidity Provision." *SIAM Journal on Financial Mathematics (forthcoming)* (2024) https://doi.org/10.1137/23M1602103.

[9] Willetts, M., Harrington, C. "Pools as Portfolios: Observed Arbitrage Efficiency and LVR Analysis of Dynamic Weight AMMs." *arXiv preprint 2602.22069* (2026) https://arxiv.org/abs/2602.22069.

[10] Martinelli, F., Mushegian, N. "A Non-Custodial Portfolio Manager, Liquidity Provider, and Price Sensor." *Balancer Labs Whitepaper* (2019) https://docs.balancer.fi/whitepaper.pdf.

[11] Baxley, B. "Maverick Protocol: Dynamic Distribution AMM." *Maverick Protocol Documentation* (2023) https://docs.mav.xyz/.

Appendix A: Contract architecture

Both phases used a Position Manager contract on Aerodrome (Base) with the following key design:

```
// Dust balance keyed by depositor and token, no pool dimension
mapping(address => mapping(address => uint256)) public dustBalance;

// Atomic rebalance: withdraw -> optional swap -> mint -> stake
function rebalanceCL(
    address pool, uint256 tokenId,
    int24 newTickLower, int24 newTickUpper,
    uint256 amount0Min, uint256 amount1Min,
    uint256 deadline,
    SwapParams swapParams
) external returns (uint256 newTokenId);
```

The critical storage layout is `dustBalance[depositor][token]`: leftover tokens from any pool's rebalance are accessible to any subsequent mint under the same depositor. This creates the cross-position channel described in §1. A vault-per-pool architecture would use `dustBalance[pool][token]` or equivalent, isolating residuals per pair.

Appendix B: Foundry test suite (reproduction)

The full suite is 15 tests across 4 contracts: 3 mock-pool tests for Theorem 1, 4 mock-pool tests for Theorem 3, 3 mock-pool tests for the directional theorems in the companion paper (Ryan, 2026b), and 5 live fork tests against Aerodrome Slipstream on Base. Requirements: Foundry `forge` (installation instructions at https://book.getfoundry.sh) and, for the fork tests, any working Base RPC URL. The `forge-std` library is included as a git submodule; the `git submodule update` command below fetches it into `foundry/lib/forge-std`.

```
cd foundry
git submodule update --init --recursive

# Mock-pool tests only, no network access required (10 tests)
```

```
forge test -vv \
  --no-match-contract '(GeometricResidualProof|DirectionalExitForkProof)$'

# Full suite, including the 5 live fork tests (15 tests)
RPC_BASE_ALCHEMY=https://mainnet.base.org forge test -vv
```

The official public endpoint above is sufficient for all five fork tests, and any Alchemy, Infura, QuickNode or other Base RPC URL is equally acceptable in `RPC_BASE_ALCHEMY`. Forge caches RPC responses under `~/.foundry/cache`, so repeat runs are fast and mock-pool tests complete in under a second. The captured `forge test -vv` output mapped to each subsection of §6 lives in `foundry/PROOF_OUTPUT.md` and is the deterministic regression target for the mock-pool tests. Fork-test residuals vary between runs as live pool state changes, but the qualitative claims (residual existence, same-range control, linear scaling, stock-NFPM non-absorption) hold deterministically.

Appendix C: Data and reproduction

The repository is at `https://github.com/g4nc0r/the-geometric-siphon`; the Zenodo Code DOI `https://doi.org/10.5281/zenodo.19905208` resolves to an immutable archival snapshot, whilst the GitHub URL tracks the latest commit. The current Foundry suite and reproduction harness in the repository align with the consolidated manuscript referenced in the supersession note. The qualitative theorem verifications described in §6 (residual existence on range changes, analytical zero on same-range rebalances, linear scaling with position size, geometric extinction without swap correction) reproduce on the current harness; specific numerical values may differ from the figures reported here, which reflect the harness used at the time of this preprint. The diffusion-event log of 1,380 rebalance events across 39 positions used for the empirical analyses in §5 is also deposited on Zenodo (Data DOI `https://doi.org/10.5281/zenodo.19904832`).